Will, Won't, Want, and Wont: A Reinforcement Learning Model of Self-control

Anonymous Author(s)

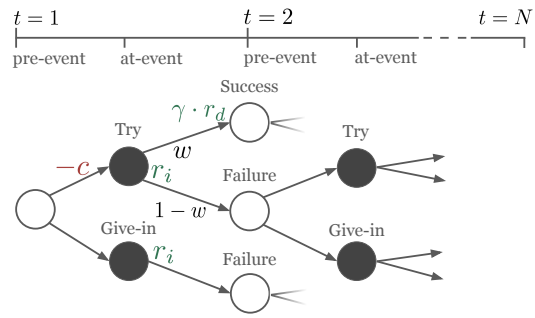
Abstract

Achieving long-term goals requires not only planning actions but also persistently executing them in the face of immediate temptations. Failures of self-control have long been studied in otherwise healthy behavior and pathological conditions such as addiction and impulsivity (Ainslie, 2001; Evenden, 1999; Rachlin, 1974; Schelling, 1984). These failures can be mitigated by commitment devices (Duckworth et al., 2018; Fudenberg and Levine, 2006), or by cognitive mechanisms collectively referred to as "willpower" (Ainslie, 2021; Loewenstein, 2000; Robinson et al., 2010). We formalize the self-control problem within a normative reinforcement learning framework in which a prospective planner decides whether to attempt resisting a temptation by accounting for its own uncertainly (in)competent willpower. The planner can improve its efficacy of exerting self-control by training its willpower. It can also enjoy an implicit determination bonus from reducing uncertainty about its degree of self-control. We assess our model's predictions when repeatedly choosing between a larger-delayed reward, and a smaller-immediate reward. Our simulations show that willpower training and exploration provide a computational account of how agents can overcome their Pavlovian biases (Dayan et al., 2006; Yee and Braver, 2018). We also show that the model captures a range of successes and failures of self-control behavior. Ultimately, explaining self-control lapses using our generative model could aid in the understanding, diagnosis, and treatment of psychiatric conditions.

Results

We consider the following paradigmatic problem of self-control: an agent with exponential discounting rate γ faces a sequence of N choices between a larger-delayed reward $\gamma \cdot r_d$ and a smaller-immediate reward r_i. For example, r_i could represent the enjoyment of eating cake while r_d represents the long-term health benefits of abstaining. Each trial consists of two stages (Fig. 1). In the "pre-event" stage, a planner chooses between "try" and "give-in". Trying to resist leads to an "at-event" stage with (willpower) probability w of success, which yields $\gamma \cdot r_d$, and probability $1-w$ of failing and receiving r_i (e.g., as a form of Pavlovian misbehaviour of the sweet-toothed; Dayan et al., 2006). Choosing giving-in deterministically yields r_i. Importantly, trying to resist incurs a cost of c, so the planner chooses to do so only when willpower is sufficiently high. In the single-trial case, the condition for try-resist is: $w \cdot \gamma \cdot r_d + (1-w) r_i - c > r_i$ (1).

The planner maximizes expected return, accounting for



Training $\quad T(w' \mid w, o) = \begin{cases} \delta(w' - \min(w+\eta_+, 1)), & \text{if } o = \text{Success} \\ \delta(w' - \max(w-\eta_-, 0)), & \text{if } o = \text{Failure} \end{cases}$

Inference $\quad p(w \mid o, h) = \dfrac{p(o \mid w) \cdot p(w \mid h)}{p(o)}, \quad \text{if } a = \text{Try}$

Integration $\quad p(w' \mid o, h) = \sum_w p(w \mid o, h) \cdot T(w' \mid w, o)$

Figure 1: **Task setting and model.** Across N trials, a pre-event decision (try-resist vs. give-in) influences an at-event outcome (if try-resist: success with probability w or failure with $1-w$). Willpower w is strengthened or weakened based on the success of resistance attempts. The planner maintains a belief over w and plans in the resulting BAMDP to prospectively account for training effects and uncertainty reduction. White and black circles are decision and action-outcome nodes, respectively. Labels on top of arrows indicate rewards (green = positive; red = negative, i.e., costs), those below indicate transition probabilities. h is the history of past observations.

both reductions in uncertainty about willpower and its potential change through training. The planner maintains a belief state $p(w)$, which is updated based on the success of resist attempts. It solves the Bayes-adaptive Markov decision process (BAMDP; Duff, 2002) using tree search. For training, willpower increases by η_+ after successful resistance, and decreases by η_- after failing to resist or giving in.

For the simulations, we set $\gamma = 1.0$, $r_i = 0.5$, $r_d = 0.8$, and $c = -0.1$, and simulate $N = 14$ trials. From equation 1, with no training and no uncertainty about w, there is a threshold $w_{thr} = \frac{1}{3}$ below which it is not worth trying to resist (Fig. 2 a-i, b-i). Here, the small chance of successfully overcoming Pavlovian impulses is not worth the cost of exerting control.

When the agent can train its willpower (training rates $\eta_+ = \eta_- = 0.2$) and can prospectively plan for the resulting changes in w, it is optimal to try to resist in early trials even below the threshold w_{thr} (Fig. 2 a-ii, b-ii). Successful resistance eventually drives w to its maximum of 1.0 (Fig. 2 d-ii, top trajectory). Repeated failures drive w down to a point where resisting is no longer worthwhile (Fig. 2 d-ii, lower two trajectories). Hence,

In submission to the *9th Conference on Cognitive Computational Neuroscience* (CCN 2026).

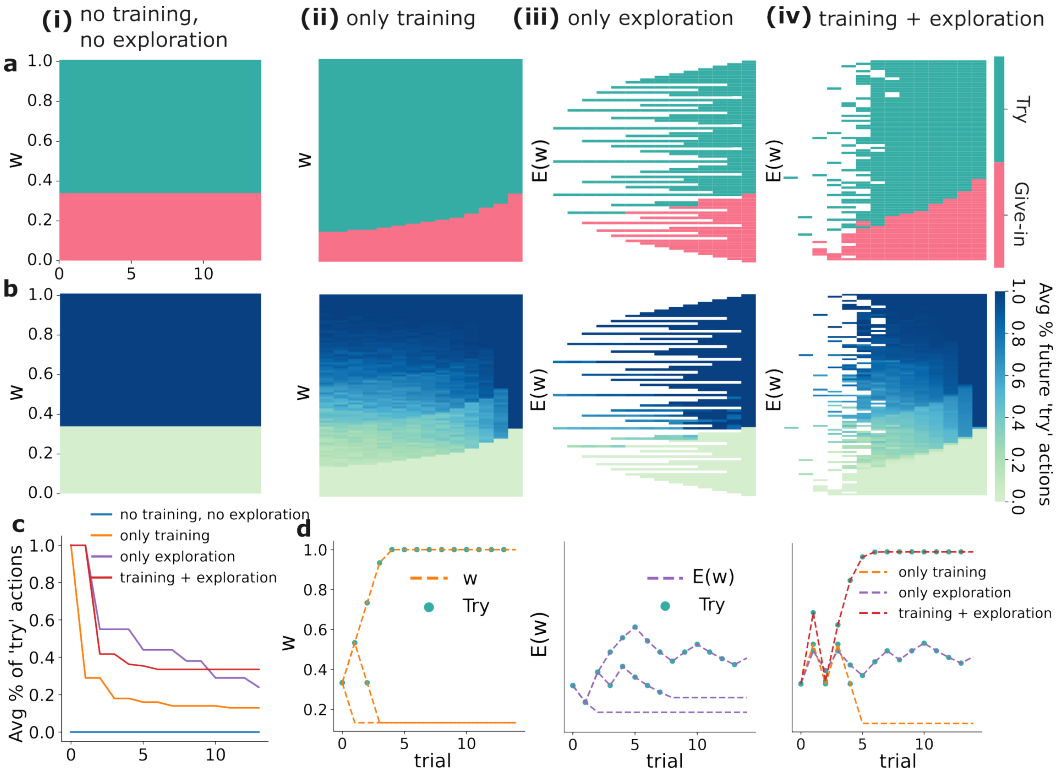


Figure 2: **Results. (a)** Optimal action ('try' vs 'give-in') given (estimated) willpower across trials. Notice that in panels ii-iv the area for trying is increased (lower than $w_{thr} = \frac{1}{3}$). For models with exploration the prior was set to Beta$(1,2)$, and whitespaces result from unreachable belief states during simulations. For training, we set $w_{init} = \frac{1}{3}$ and for uncertainty, $w_{real} = \frac{1}{3}$. **(b)** Percentage of future 'try' actions across 100 realizations given w or $\mathbb{E}(w)$ and t. **(c)** Average proportion of 'try' actions across 100 realizations for all conditions. **(d)** Sample trajectories for ii) 'only training', iii) 'only exploration', and iv) comparing them to 'training + exploration'.

current behavior can be seen as setting 'precedents' for future trials, with present actions coming to influence future ones, effectively 'bundling' trials together (Ainslie, 2001, 2021).

When the agent is uncertain about its willpower but cannot train it, there is still an incentive to try to resist in early trials even when $\mathbb{E}(w) < w_{thr}$ (Fig. 2 a-iii, b-iii). This is because trying to resist provides an information gain that we call a determination bonus. If the agent successfully resists in early trials, the resulting increase in $\mathbb{E}(w)$ can sustain continued resistance (Fig. 2 d-iii, top trajectory); however, if there were enough trials, the agent would eventually learn its true willpower. Conversely, failures – even unfortunate ones – drive $\mathbb{E}(w)$ downward, potentially leading to self-censoring in which the agent never discovers its true willpower (Fig. 2 d-iii, lower two trajectories).

Finally, exploration and training can reinforce each other. In Fig. 2 d-iv, we contrast example trajectories on the same problem. Initial failures lead the "only training" model to give up and not try again. The "only exploration" model persists but willpower remains at an intermediate level. In contrast, the "training + exploration" model persists, and successfully drives willpower to its maximum. On average, there is more self-control when either or both mechanisms play a role than without them (Fig. 2 c).

Discussion

Our simulations capture behavior characteristic of both failures and successes of self-control. The agent is motivated to try resistance by both uncertainty about, and the prospect of increasing, its willpower. However, these factors only govern the decision to try; the long-term outcome is stochastic, and repeated failures reduce inferred (or actual) willpower, eventually leading the agent to stop trying altogether. Conversely, with training, repeated successes drive willpower to its maximum.

In our model, self-control failure arises from a Pavlovian bias to defect – a mechanism that can be related to hyperbolic discounting as suggested by Ainslie, 2001; Ainslie and Herrnstein, 1981. With training, successful resistance strengthens self-control; this has several psychological interpretations and links, including practice-dependent cognitive control (Jones et al., 2016), precedent-setting (Ainslie, 2001, 2021; Ainslie and Monterosso, 2003), and habit formation (Gershman, 2020). Unlike Bénabou and Tirole, 2004, which model persistent behavior as a form of self-deception in which the agent maintains false beliefs about high willpower, our mechanisms reflect changes to automatic control processes governing resistance. They can be interpreted in terms of intrapersonal trust or bundling (Ainslie, 2021; Ainslie and Monterosso, 2003).

For healthy self-control, excessive restraint is also undesirable; it must be balanced with selective indulgence under exceptional circumstances. In future work, we plan to extend our model to account for such exception trials where defection may be rational, and investigate how the desire to protect willpower can lead to compulsion—resisting even during exceptions. We will also investigate maladaptive behaviors such as relapses (Brandon et al., 2007), which may arise from asymmetric training rates in which failures reduce willpower more than successes restore it, increasing the likelihood that an unlucky sequence of failures drives willpower toward zero.

References

Ainslie, G. (2001). *Breakdown of will*. Cambridge University Press.

Ainslie, G. (2021). Willpower with and without effort. *Behavioral and Brain Sciences*, *44*, e30.

Ainslie, G., & Herrnstein, R. J. (1981). Preference reversal and delayed reinforcement. *Animal Learning & Behavior*, *9*(4), 476–482.

Ainslie, G., & Monterosso, J. (2003). Will as intertemporal bargaining: Implications for rationality. *University of Pennsylvania Law Review*, *151*(3), 825–862.

Bénabou, R., & Tirole, J. (2004). Willpower and personal rules. *Journal of Political Economy*, *112*(4), 848–886.

Brandon, T. H., Vidrine, J. I., & Litvin, E. B. (2007). Relapse and relapse prevention. *Annu. Rev. Clin. Psychol.*, *3*, 257–284.

Dayan, P., Niv, Y., Seymour, B., & Daw, N. D. (2006). The misbehavior of value and the discipline of the will. *Neural networks*, *19*(8), 1153–1160.

Duckworth, A. L., Milkman, K. L., & Laibson, D. (2018). Beyond willpower: Strategies for reducing failures of self-control. *Psychological Science in the Public Interest*, *19*(3), 102–129.

Duff, M. O. (2002). *Optimal learning: Computational procedures for bayes-adaptive markov decision processes*. University of Massachusetts Amherst.

Evenden, J. L. (1999). Varieties of impulsivity. *Psychopharmacology*, *146*(4), 348–361.

Fudenberg, D., & Levine, D. K. (2006). A dual-self model of impulse control. *American economic review*, *96*(5), 1449–1476.

Gershman, S. J. (2020). Origin of perseveration in the trade-off between reward and complexity. *Cognition*, *204*, 104394.

Jones, A., Di Lemma, L. C., Robinson, E., Christiansen, P., Nolan, S., Tudur-Smith, C., & Field, M. (2016). Inhibitory control training for appetitive behaviour change: A meta-analytic investigation of mechanisms of action and moderators of effectiveness. *Appetite*, *97*, 16–28.

Loewenstein, G. (2000). Willpower: A decision-theorist's perspective. *Law and Philosophy*, 51–76.

Rachlin, H. (1974). Self-control. *Behaviorism*, *2*(1), 94–107.

Robinson, M. D., Schmeichel, B. J., & Inzlicht, M. (2010). A cognitive control perspective of self-control strength and its depletion. *Social and Personality Psychology Compass*, *4*(3), 189–200.

Schelling, T. C. (1984). Self-command in practice, in policy, and in a theory of rational choice. *The American Economic Review*, *74*(2), 1–11.

Yee, D. M., & Braver, T. S. (2018). Interactions of motivation and cognitive control. *Current opinion in behavioral sciences*, *19*, 83–90.